



ELMALG: Linear Equations and Inequalities

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Learning Objectives

- Solve one- and two-step linear equations
- Solve linear inequalities and graph the solution on a number line
- Apply linear equations to real-world word problems
- Check solutions by substituting back into the equation

Simplify each expression completely. Show all steps and circle your final answer.

Linear inequalities

1. Solve the inequality: $1x + 1 < 13$. Express x as a fraction if needed.

$$1x + 1 < 13$$

Answer: _____

2. Solve and graph the solution: $1x + 5 > 9$

$$1x + 5 > 9$$

Answer: _____

3. Solve: $4x + 9 \leq 22$. Write the solution in interval notation.

$$4x + 9 \leq 22$$

Answer: _____

4. Solve the inequality: $2x + 9 < 22$. Express x as a fraction if needed.

$$2x + 9 < 22$$

Answer: _____

5. Solve and graph the solution: $3x + 1 > 21$

$$3x + 1 > 21$$

Answer: _____

6. Solve: $5x + 4 \leq 19$. Write the solution in interval notation.

$$5x + 4 \leq 19$$

Answer: _____



7. Solve the inequality: $3x + 5 < 23$. Express x as a fraction if needed.

$$3x + 5 < 23$$

Answer: _____

8. Solve and graph the solution: $4x + 3 > 22$

$$4x + 3 > 22$$

Answer: _____

9. Solve: $5x + 3 \leq 7$. Write the solution in interval notation.

$$5x + 3 \leq 7$$

Answer: _____

10. Solve the inequality: $1x + 2 < 10$. Express x as a fraction if needed.

$$1x + 2 < 10$$

Answer: _____

11. Solve and graph the solution: $1x + 5 > 19$

$$1x + 5 > 19$$

Answer: _____

12. Solve: $3x + 9 \leq 24$. Write the solution in interval notation.

$$3x + 9 \leq 24$$

Answer: _____

Solving two-step equations

13. Solve for x : $3x + 5 = 5$.

$$3x + 5 = 5$$

Answer: _____

14. A plumber charges \$48 per hour plus a flat \$24 fee. If the total bill is \$202, how many hours did the job take? Solve $48x + 24 = 202$.

$$48x + 24 = 202$$

Answer: _____



15. Solve for x: $4x + -8 = 8$. Check your answer.

$$4x + -8 = 8$$

Answer: _____

16. Solve for x: $2x + 6 = 7$.

$$2x + 6 = 7$$

Answer: _____

17. A plumber charges \$22 per hour plus a flat \$32 fee. If the total bill is \$138, how many hours did the job take? Solve $22x + 32 = 138$.

$$22x + 32 = 138$$

Answer: _____

18. Solve for x: $8x + 7 = -5$. Check your answer.

$$8x + 7 = -5$$

Answer: _____

19. Solve for x: $4x + 6 = -1$.

$$4x + 6 = -1$$

Answer: _____

20. A plumber charges \$35 per hour plus a flat \$27 fee. If the total bill is \$261, how many hours did the job take? Solve $35x + 27 = 261$.

$$35x + 27 = 261$$

Answer: _____

21. Solve for x: $5x + -1 = 4$. Check your answer.

$$5x + -1 = 4$$

Answer: _____

Variables on both sides

22. Solve for x: $7x + 6 = 2x + 14$

$$7x + 6 = 2x + 14$$

Answer: _____



23. Two friends are saving money. Alex has \$42 and saves \$6 per week. Jordan has \$12 and saves \$23 per week. After how many weeks will they have the same amount?

$$6x + 42 = 23x + 12$$

Answer: _____

24. Solve for x: $12x + 3 = 3x + 30$. (Hint: $m_1 > m_2$ here, so collect x terms on the left.)

$$12x + 3 = 3x + 30$$

Answer: _____

25. Solve for x: $3x + 6 = 5x + 5$

$$3x + 6 = 5x + 5$$

Answer: _____

26. Two friends are saving money. Alex has \$32 and saves \$14 per week. Jordan has \$17 and saves \$16 per week. After how many weeks will they have the same amount?

$$14x + 32 = 16x + 17$$

Answer: _____

27. Solve for x: $8x + 1 = 4x + 10$. (Hint: $m_1 > m_2$ here, so collect x terms on the left.)

$$8x + 1 = 4x + 10$$

Answer: _____

28. Solve for x: $3x + 5 = 4x + 12$

$$3x + 5 = 4x + 12$$

Answer: _____

29. Two friends are saving money. Alex has \$27 and saves \$6 per week. Jordan has \$34 and saves \$23 per week. After how many weeks will they have the same amount?

$$6x + 27 = 23x + 34$$

Answer: _____

30. Solve for x: $12x + 6 = 1x + 29$. (Hint: $m_1 > m_2$ here, so collect x terms on the left.)

$$12x + 6 = 1x + 29$$

Answer: _____





ELMALG: Linear Equations and Inequalities

ANSWER KEY & SOLUTIONS

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Topics: Linear inequalities, Variables on both sides, Solving two-step equations. All answers verified by independent computation.

Solutions



Linear inequalities

1. Solve the inequality: $1x + 1 < 13$. Express x as a fraction if needed.

$$1x + 1 < 13$$

→ Subtract 1: $1x < 12$.

→ Divide by 1: $x < 12$.

Answer: $x < 12$

2. Solve and graph the solution: $1x + 5 > 9$

$$1x + 5 > 9$$

→ Subtract 5: $1x > 4$.

→ Divide by 1: $x > 4$.

→ Open circle at 4, shade right.

Answer: $x > 4$

3. Solve: $4x + 9 \leq 22$. Write the solution in interval notation.

$$4x + 9 \leq 22$$

→ Subtract 9: $4x \leq 13$.

→ Divide by 4: $x \leq 13/4$.

→ Closed circle at $13/4$, shade left. Interval: $(-\infty, 13/4]$.

Answer: $x \leq \frac{13}{4}$

4. Solve the inequality: $2x + 9 < 22$. Express x as a fraction if needed.

$$2x + 9 < 22$$

→ Subtract 9: $2x < 13$.

→ Divide by 2: $x < 13/2$.

Answer: $x < \frac{13}{2}$

5. Solve and graph the solution: $3x + 1 > 21$

$$3x + 1 > 21$$

→ Subtract 1: $3x > 20$.

→ Divide by 3: $x > 20/3$.

→ Open circle at $20/3$, shade right.

Answer: $x > \frac{20}{3}$

6. Solve: $5x + 4 \leq 19$. Write the solution in interval notation.

$$5x + 4 \leq 19$$

→ Subtract 4: $5x \leq 15$.

→ Divide by 5: $x \leq 3$.

→ Closed circle at 3, shade left. Interval: $(-\infty, 3]$.

Answer: $x \leq 3$



7. Solve the inequality: $3x + 5 < 23$. Express x as a fraction if needed.

$$3x + 5 < 23$$

→ Subtract 5: $3x < 18$.

→ Divide by 3: $x < 6$.

Answer: $x < 6$

8. Solve and graph the solution: $4x + 3 > 22$

$$4x + 3 > 22$$

→ Subtract 3: $4x > 19$.

→ Divide by 4: $x > 19/4$.

→ Open circle at $19/4$, shade right.

Answer: $x > \frac{19}{4}$

9. Solve: $5x + 3 \leq 7$. Write the solution in interval notation.

$$5x + 3 \leq 7$$

→ Subtract 3: $5x \leq 4$.

→ Divide by 5: $x \leq 4/5$.

→ Closed circle at $4/5$, shade left. Interval: $(-\infty, 4/5]$.

Answer: $x \leq \frac{4}{5}$

10. Solve the inequality: $1x + 2 < 10$. Express x as a fraction if needed.

$$1x + 2 < 10$$

→ Subtract 2: $1x < 8$.

→ Divide by 1: $x < 8$.

Answer: $x < 8$

11. Solve and graph the solution: $1x + 5 > 19$

$$1x + 5 > 19$$

→ Subtract 5: $1x > 14$.

→ Divide by 1: $x > 14$.

→ Open circle at 14, shade right.

Answer: $x > 14$

12. Solve: $3x + 9 \leq 24$. Write the solution in interval notation.

$$3x + 9 \leq 24$$

→ Subtract 9: $3x \leq 15$.

→ Divide by 3: $x \leq 5$.

→ Closed circle at 5, shade left. Interval: $(-\infty, 5]$.

Answer: $x \leq 5$



Solving two-step equations

13. Solve for x : $3x + 5 = 5$.

$$3x + 5 = 5$$

→ Subtract 5 from both sides: $3x = 0$.

→ Divide by 3: $x = 0$.

Answer: $x = 0 \div 3 = 0$

14. A plumber charges \$48 per hour plus a flat \$24 fee. If the total bill is \$202, how many hours did the job take?
Solve $48x + 24 = 202$.

$$48x + 24 = 202$$

→ Subtract \$24: $48x = 178$. Divide by 48: $x = 89/24$ hours.

Answer: $x = 178 \div 48 = 89/24$

15. Solve for x : $4x + -8 = 8$. Check your answer.

$$4x + -8 = 8$$

→ Subtract -8 from both sides: $4x = 16$.

→ Divide both sides by 4: $x = 4$.

Answer: $x = 16 \div 4 = 4$

16. Solve for x : $2x + 6 = 7$.

$$2x + 6 = 7$$

→ Subtract 6 from both sides: $2x = 1$.

→ Divide by 2: $x = 1/2$.

Answer: $x = 1 \div 2 = 1/2$

17. A plumber charges \$22 per hour plus a flat \$32 fee. If the total bill is \$138, how many hours did the job take?
Solve $22x + 32 = 138$.

$$22x + 32 = 138$$

→ Subtract \$32: $22x = 106$. Divide by 22: $x = 53/11$ hours.

Answer: $x = 106 \div 22 = 53/11$

18. Solve for x : $8x + 7 = -5$. Check your answer.

$$8x + 7 = -5$$

→ Subtract 7 from both sides: $8x = -12$.

→ Divide both sides by 8: $x = -3/2$.

Answer: $x = -12 \div 8 = -3/2$



19. Solve for x: $4x + 6 = -1$.

$$4x + 6 = -1$$

→ Subtract 6 from both sides: $4x = -7$.

→ Divide by 4: $x = -7/4$.

Answer: $x = -7 \div 4 = -7/4$

20. A plumber charges \$35 per hour plus a flat \$27 fee. If the total bill is \$261, how many hours did the job take?
Solve $35x + 27 = 261$.

$$35x + 27 = 261$$

→ Subtract \$27: $35x = 234$. Divide by 35: $x = 234/35$ hours.

Answer: $x = 234 \div 35 = 234/35$

21. Solve for x: $5x + -1 = 4$. Check your answer.

$$5x + -1 = 4$$

→ Subtract -1 from both sides: $5x = 5$.

→ Divide both sides by 5: $x = 1$.

Answer: $x = 5 \div 5 = 1$



Variables on both sides

22. Solve for x : $7x + 6 = 2x + 14$

$$7x + 6 = 2x + 14$$

→ Move x terms left: $7x - 2x = 14 - 6$.

→ Combine: $5x = 8$.

→ Divide by 5: $x = 8/5$.

Answer: $x = \frac{8}{5}$

23. Two friends are saving money. Alex has \$42 and saves \$6 per week. Jordan has \$12 and saves \$23 per week. After how many weeks will they have the same amount?

$$6x + 42 = 23x + 12$$

→ Set equations equal: $6x + 42 = 23x + 12$.

→ Move x terms: $6x - 23x = 12 - 42$. So $-17x = -30$.

→ $x = 30/17$ weeks.

Answer: $x = \frac{30}{17}$

24. Solve for x : $12x + 3 = 3x + 30$. (Hint: $m_1 > m_2$ here, so collect x terms on the left.)

$$12x + 3 = 3x + 30$$

→ Subtract $3x$ from both sides: $(12 - 3)x + 3 = 30$.

→ Subtract 3: $9x = 27$.

→ Divide by 9: $x = 3$.

Answer: $x = 3$

25. Solve for x : $3x + 6 = 5x + 5$

$$3x + 6 = 5x + 5$$

→ Move x terms left: $3x - 5x = 5 - 6$.

→ Combine: $-2x = -1$.

→ Divide by -2 : $x = 1/2$.

Answer: $x = \frac{1}{2}$

26. Two friends are saving money. Alex has \$32 and saves \$14 per week. Jordan has \$17 and saves \$16 per week. After how many weeks will they have the same amount?

$$14x + 32 = 16x + 17$$

→ Set equations equal: $14x + 32 = 16x + 17$.

→ Move x terms: $14x - 16x = 17 - 32$. So $-2x = -15$.

→ $x = 15/2$ weeks.

Answer: $x = \frac{15}{2}$



27. Solve for x: $8x + 1 = 4x + 10$. (Hint: $m_1 > m_2$ here, so collect x terms on the left.)

$$8x + 1 = 4x + 10$$

→ Subtract $4x$ from both sides: $(8 - 4)x + 1 = 10$.

→ Subtract 1: $4x = 9$.

→ Divide by 4: $x = 9/4$.

Answer: $x = \frac{9}{4}$

28. Solve for x: $3x + 5 = 4x + 12$

$$3x + 5 = 4x + 12$$

→ Move x terms left: $3x - 4x = 12 - 5$.

→ Combine: $-1x = 7$.

→ Divide by -1: $x = -7$.

Answer: $x = -7$

29. Two friends are saving money. Alex has \$27 and saves \$6 per week. Jordan has \$34 and saves \$23 per week. After how many weeks will they have the same amount?

$$6x + 27 = 23x + 34$$

→ Set equations equal: $6x + 27 = 23x + 34$.

→ Move x terms: $6x - 23x = 34 - 27$. So $-17x = 7$.

→ $x = -7/17$ weeks.

Answer: $x = \frac{-7}{17}$

30. Solve for x: $12x + 6 = 1x + 29$. (Hint: $m_1 > m_2$ here, so collect x terms on the left.)

$$12x + 6 = 1x + 29$$

→ Subtract $1x$ from both sides: $(12 - 1)x + 6 = 29$.

→ Subtract 6: $11x = 23$.

→ Divide by 11: $x = 23/11$.

Answer: $x = \frac{23}{11}$

